Application of STEM principles and practices

The proposed solution is well-substantiated with STEM principles and practices applicable to all or nearly all design requirements and functional claims; there is substantial evidence that the application of those principles and practices by the student or a suitable alternate has been reviewed by two or more experts (qualified consultants and/or project mentors) and that those reviews provide confirmation (verification) or detail necessary to inform a corrective response.

Using STEM Principles and Expert Feedback in the Design

Our project focuses on designing and building a steel frame for lifting large animals safely. The goal is to raise animals that weigh up to one ton. To do this, we followed important STEM ideas to make sure our design was strong, safe, and able to do the job. We also tested the frame and got feedback from experts to make sure everything works correctly and meets safety needs.

How STEM Ideas Helped Our Design

We started by choosing steel as our main material. Steel is strong and doesn't bend easily, which makes it a good choice for heavy lifting. Engineers use steel in buildings, bridges, and machines, so we knew it could work well for our frame. We made sure the frame was built in a way that could hold a lot of weight without breaking or twisting.

Our design uses a strong base with vertical, horizontal, and diagonal pieces of steel to keep it stable. We thought carefully about how the weight would be spread out across the frame, and we looked at the forces that would push or pull on different parts. We used basic engineering knowledge, like how forces move through a structure, to decide where to place each steel beam. We also added extra strength where needed to make sure the frame wouldn't fail under pressure.

Testing the Frame

To prove the frame could actually support the weight, we hung a one-ton load from it for several minutes. The frame stayed strong and didn't bend or break. This real test showed that the frame worked just like we planned. We also moved the weight a little during the test to see if the frame could handle shifting loads, which might happen if the animal moves. Everything stayed stable and held up well.

This is all of us hanging from the frame and it stays perfectly stable. Combined we weigh about 750 pounds.

This is the tarp that will go under and attach to the frame we have built. This tarp is sturdy and able to hold the weight of the animals. The zookeepers have used this in the past and it has worked successfully. This tarp paired with our frame should lead to a successful product for the Zoo.

Using Engine Levelers to Keep Balance

To keep the animal steady while it's being lifted, we added two engine levelers—one on each side of the frame. These tools are usually used to lift and balance car engines, but we used them to help spread the animal's weight evenly. They allow the person operating the lift to adjust the angle of the animal as it goes up, making sure it stays level and safe.

This setup helps balance the load so that one side doesn't carry more weight than the other. It also keeps the animal from swinging or tipping during the lift. This is important because uneven weight could cause stress on the frame or hurt the animal.

Making the Lift Safe and Ethical for Animals

Because we're lifting live animals, we worked hard to make sure the design is safe and respectful. We made sure there are no sharp edges and the surfaces are stable. We followed common safety rules used in engineering and added extra strength in our design to handle surprises, like sudden movement or off-center weight.

Experts Helped Improve the Design

We didn’t just rely on our own knowledge—we asked for help from people with more experience. One expert who helped us was Stacy Gardener, an engineer and mentor who understands mechanical systems. She looked at our frame after we built it and watched it during the weight test. She said it was built well and gave us good advice, like making sure our welds were solid and adding braces in a few areas to make it even stronger. We followed her suggestions and made changes to improve the frame.

We also had help from our teachers and mentors at school who know about engineering and building projects. They looked at our plans, watched us build, and said the engine levelers were a good choice for keeping the load balanced. They gave us tips to make the design even better, like adding safety stops or stronger hooks, which we might include in the next version.

Finally, our project shows how STEM knowledge can solve real problems. We used engineering and math to design a safe, strong frame, tested it to prove it works, and improved it with expert advice. We feel confident that our animal lift meets all the goals we started with and is a safe and useful tool for lifting large animals. By following the design process and listening to experts, we created something that works well in the real world.